

Lecture 07

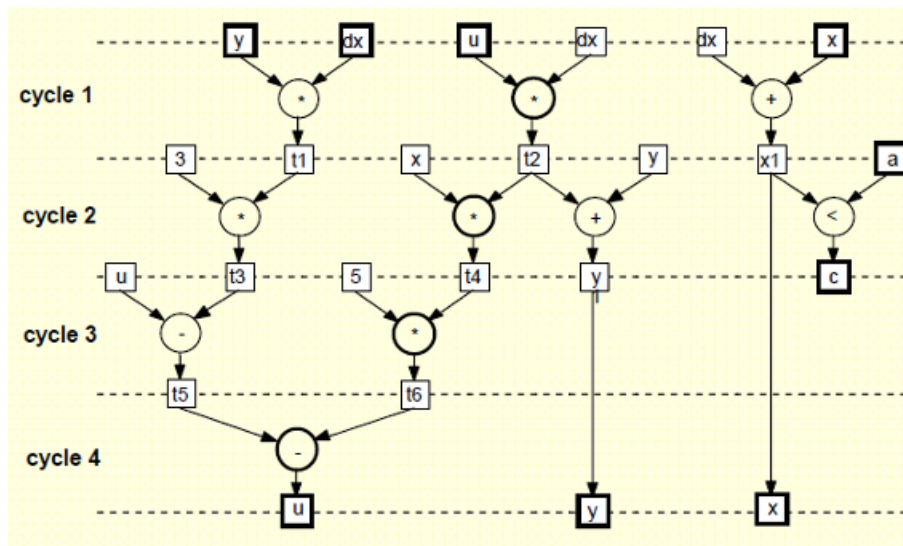
High Level Hardware Synthesis II

Descriptive Question

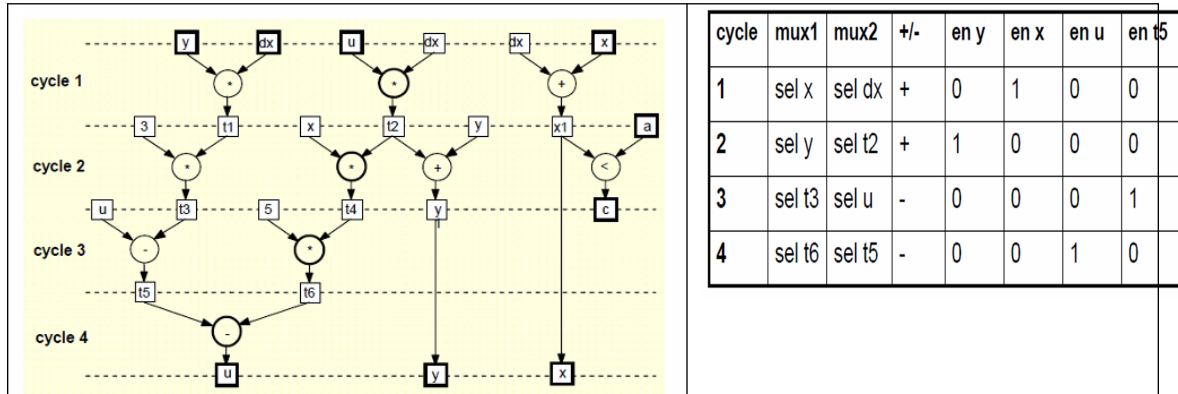
1. Explain resource allocation in HLS.
2. Explain binding, and how it differs from allocation.
3. Explain datapath synthesis in High-Level Synthesis.
4. Explain the main components of a synthesized datapath.
5. Describe the main functions performed by a controller in HLS.
6. Describe chaining in High-Level Synthesis.
7. Describe multicycling, and why it is used.

Mathematical Question

1. Consider the following Data Flow Graph (DFG) and answer the following questions:

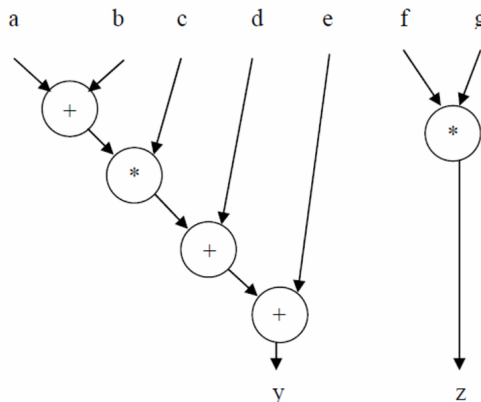


- a) Perform Allocation from the above DFG.
 - b) Perform Binding from the above DFG.
 - c) Construct a datapath from the DFG.
2. Consider the following control table and DFG and answer the following questions:



- Convert the control table into finite State Machine (FSM).
- Construct a complete datapath from the FSM.
- Build a complete control table.

3. Consider the following data flow graph:



The following resources are available:

- one adder which requires 40 ns per addition,
- one multiplier which requires 80 ns per operation.
- We use multicycling to achieve a cycle time of 50 ns.

- Perform a list scheduling of the data flow graph under the above assumptions.
- Try to find a better schedule with the same resources but less latency.

(Practice the questions discussed in the class Lecture)